



IS 200 - SYSTEMS ANALYSIS AND DESIGN

GROUP ASSIGNMENT COURSEWORK

YEAR 2 SEMESTER 1

REQUIREMENTS GATHERING REPORT

PROPOSED STUDENT MANAGENT SYSTEM (SMS)

FOR

WOMEN'S INSTITUTE OF TECHNOLOGY & INNOVATION

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BY

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1. INTRODUCTION

1.1 Background

The existing file-based approach at the Women's Institute of Technology and Innovation has become inadequate for effectively managing student and administrative data. Relying on manual processes leads to challenges such as delays in accessing information, difficulties in tracking attendance, and complications in managing student records and results. The proposed Student Management System (SMS) seeks to resolve these issues by offering a streamlined digital solution to enhance both academic and administrative operations.

1.2 Purpose

The purpose of this field study report is to present findings, analyze requirements, and outline the functional and non-functional specifications for the development of a Student Management System (SMS) for Women's Institute of Technology and Innovation. This system aims to automate student record management and streamline attendance tracking.

The Student Management System will replace paper-based and file-based student record tracking with a centralized digital solution. It is designed to improve accuracy, eliminate data redundancy, and enhance accessibility and operational efficiency through automated QR code attendance and digital record management.

1.3 Scope of the Study

The system will cover essential areas of the student lifecycle, including student application, enrollment, course management, attendance tracking, assessment, grade management, and reporting. Through the SMS, students, lecturers, and administrative staff will be able to access information and perform necessary tasks efficiently.

2. REQUIREMENTS GATHERING

2.1 Objective of Requirement Gathering

The objective of requirement gathering for the Student Management System (SMS) is to fully understand user needs, define essential functionalities, and establish standards for security, usability,

and performance. This process also identifies any technical or organizational constraints and sets business rules to ensure data integrity and proper access, creating a system that supports the institution's academic and administrative needs effectively.

2.2 Methodology for Requirement Gathering

The primary requirements gathering instrument was interviews with guided questions. This format enabled us to capture in-depth responses, allowing participants to elaborate on their experiences and expectations for the SMS. The interview questions covered the following areas;

1. Desired functionalities in the SMS.
2. Challenges with the current file-based system.
3. Expected roles and tasks for different user types.
4. Types of student information the SMS should manage.
5. Specific reporting and data insights required.

2.3 Participants

We collected data from a total of 15 participants, including students, lecturers, and administrative staff, who are primary users of the system. Their responses provided valuable input for defining system requirements.

2.3 Data Collection Process

Interviews were conducted in person to encourage detailed responses and allow interviewers to clarify questions as needed. Responses were documented to ensure accurate analysis and insights for each key area.

3. DATA ANALYSIS

This section summarizes the feedback gathered during the interviews, highlighting the essential requirements and functionalities that the SMS should incorporate.

3.1 Desired Functionalities in the SMS

Participants identified several crucial functionalities, including:

- **Student Application** : Simplifying and digitizing the application process for new students.

- **Academic Resources:** Access to lecture notes, assignments, exam schedules, and resources such as textbooks.
- **Centralized Academic Calendar:** Displaying important dates, including assignment deadlines and exam schedules.
- **Payment Processing:** Tracking tuition fees, generating receipts, and managing outstanding balances.
- **Attendance Management:** A QR code scanning feature for recording attendance accurately and efficiently.
- **Grade Assessment and Feedback:** Storing assessment records with lecturers' comments and feedback.
- **Reporting and Analytics:** Generating reports for attendance, grades, and performance tracking.
- **Notifications and Alerts:** The system should send notifications students on each system event. The system should notify facilitators on all system activity events.
- **Data Export:** The system should allow both administrators and facilitators to export attendance and grade data in various formats (e.g., PDF, Excel).

3.2 Desired Approach for Attendance Management

Participants emphasized the need for an efficient QR code-based attendance management system.

The suggested approach involves:

1. **QR Code Generation:** Each class or event would have a unique QR code generated by the system.
2. **Attendance Logging:** Students scan the code with their devices when attending classes, which logs their attendance in real time.
3. **Tracking and Reporting:** The system maintains a record of attendance logs, enabling lecturers and administrative staff to view daily or cumulative attendance summaries and address any inconsistencies.

This approach is expected to save time, reduce inaccuracies, and simplify attendance tracking, especially compared to manual entry methods.

3.3 Challenges with the Current File-Based Approach

Participants cited several issues with the existing file-based system:

- Slow data retrieval and delays in accessing student records.
- Inefficiencies in manually updating student information.
- Challenges in tracking attendance accurately.
- Limited capabilities for generating reports on student progress and attendance.

3.4 Expected Roles and Tasks for Users

- **Students:** Registration, course enrollment, attendance tracking, and accessing grades and feedback.
- **Lecturers:** Uploading and managing assignments, tests, exams, and tracking student performance.
- **Administrative Staff:** Overseeing student records, processing enrollments, managing payment records, and generating reports.

3.5 Student Application Flow

Participants highlighted the need for a streamlined student application process, which could involve:

1. **Application Submission:** Prospective students submit their applications via the company website, uploading necessary documents and filling in required information.
2. **Application Review:** Administrative staff reviews the applications, checks for eligibility, and flags any incomplete submissions.
3. **Approval and Enrollment:** Approved applications move forward in the process, and students are officially enrolled into the system. An automated notification confirms their enrollment status and provides them with access credentials.
4. **Tracking and Follow-Up:** The system allows prospective students and administrative staff to track application status and follow up as necessary.

This flow aims to improve the application experience, reduce paperwork, and provide a clear tracking mechanism for both applicants and staff.

3.6 Information Management Requirements

Participants emphasized that the SMS should be able to:

- Maintain comprehensive records on each student, including personal information, academic progress, and attendance.

- Provide a user-friendly interface for storing and retrieving student-related data.

3.7 Reporting and Data Insights

Participants expressed the need for regular, accessible reports, including:

- Attendance summaries to monitor student engagement.
- Grade distribution and academic performance reports.
- Enrollment and payment tracking for administrative analysis.

4. System Requirements

4.1 Functional Requirements

Based on interview responses, the following functional requirements have been identified for the SMS:

1. User Management

- Define roles (Student, Lecturer, Admin) with specific access permissions.
- Manage user profiles, including secure login credentials and role-based access.

2. Student Information Management

- Store and manage essential student details, such as ID, contact information, and academic records.
- Enable easy access to attendance records, grades, and feedback from lecturers.

3. Enrollment and Registration

- Allow students to register for courses digitally.
- Enable administrative oversight of course enrollment status.

4. Course Management

- Manage course information, including objectives, outcomes, and content.
- Allow lecturers to update course-related materials and assessments.

5. Attendance Tracking

- Provide QR code scanning for quick attendance logging.
- Track attendance and generate reports on student engagement.

6. **Student Application**

- Streamline the application process with an online submission, review, and tracking mechanism.
- Allow administrative staff to review applications, manage enrollment statuses, and notify applicants.

7. **Academic Calendar**

- Maintain a centralized calendar with key dates, such as assignment deadlines and exam schedules.
- Allow updates by administrative staff for real-time access.

8. **Payment Processing**

- Track tuition and fee payments, with receipt generation and reporting.
- Enable administrators to monitor payment records for financial management.

9. **Grading and Assessment**

- Allow lecturers to input grades and provide feedback.
- Enable students to view grades and comments in real time.

10. **Reports and Analytics**

- Generate customized reports on attendance, grades, and course progress.
- Provide analytics to aid in academic and administrative decision-making.

4.2 Non-Functional Requirements

These requirements specify the expected performance, security, and usability standards of the SMS.

1. **Usability:** The system interface should be user-friendly, with simple navigation for different roles (students, facilitators, administrators). The system should be accessible across devices, including desktops, tablets, and smartphones. QR code scanning shall be efficient, requiring minimal interaction for attendance capture.

2. **Reliability:** The system should function consistently, with minimal downtime and robust data backup mechanisms.
3. **Performance:** The system must process data quickly to prevent delays in retrieving or updating information.
4. **Security:** Role-based and permission access control alongside user authentication are required to protect sensitive data. The system should use secure logins with password protection for all user roles. All data transmissions, including QR code data, should be encrypted. The system should enforce confidentiality.
5. **Scalability:** The system should be capable of handling increasing numbers of users and data entries without impacting performance.
6. **Data Integrity:** Ensure data is accurate and modifications are authorized.
7. **Compliance:** Adhere to relevant data protection regulations, such as GDPR, to safeguard user data.

5.4 Business Rules

The following business rules define operational principles and integrity constraints for the WITI Student Management System (SMS). These rules outline data relationships and access restrictions to ensure system integrity and enforce specific requirements for managing students, staff, and academic programs.

Integrity Constraints on Data

1. Department and Institute Relationships

- Every department added must belong to the institute.

2. Role and Permission Management

- The system can have one or more roles.
- Each permission must belong to a role.
- A role can have multiple users assigned to it.

3. User Data Integrity

- Each user must have a unique email and phone number.
- A user must have an assigned role.

4. Program Structure

- A program is limited to two semesters per year.
- Each program can have one or more cohorts.
- Programs must have unique names to avoid confusion.

5. Course and Semester Organization

- A course must be assigned to a specific semester.
- Each semester can include one or more courses.
- Each semester must have a unique name.
- Each course must have a unique name.

6. Cohort and Student Associations

- Cohorts must have unique names.
- Each cohort can contain multiple students.
- Every student must have a unique registration number.
- A student can only belong to one cohort

7. Facilitator and Course Assignments

- Each facilitator can only teach one course.

8. Application Process

- A new student can only apply once for a specific course for every intake.
- Applications are tied exclusively to individual students.
- An application form can include one or more referral sources.
- There can be one or more referral sources.

9. Student Enrollment and Cohort Requirements

- Every student must be assigned to a particular cohort.
- Students must enroll in each new semester.
- A student can enroll in multiple courses (course units) within a given semester.

10. Grades Assessment

- Coursework and tests contribute 40% of the final grade, while the final exam contributes 60%.
- Students scoring below 50% must retake the course unit with a fee of UGX 30,000
- Each grade entry must link to a valid student and course.
- Grades below 50% automatically trigger a retake requirement.
- Each lecturer can only add marks for the course he or she teaches.
- Students can comment on their marks scored
- Each student can access only their student details and scored marks.

11. Attendance with QR Code requirements

- Students must scan a QR code to mark attendance, verifying presence at the scheduled time and location.
- Attendance contributes to the final grade, with thresholds for minimum required attendance (10%) for course completion.
- Each QR code is unique to the session and expires after a set time to prevent misuse.
- QR codes are linked to specific courses and students, ensuring valid attendance records.
- Only students physically present can mark attendance to avoid proxy check-ins.

12. Financial Obligations

- A student must pay at least half of their tuition to gain access to the system.
- Some students are given scholarships.

CONCLUSION

The field study provided key insights into the requirements for the Student Management System (SMS). Participants highlighted essential features such as student data management, assignment tracking, and attendance solutions, while noting challenges in the current file-based system, like inefficiency and limited accessibility. Clear user roles and tasks for students, lecturers, and staff were identified, along with a need for regular reporting and data insights. These findings form a

solid foundation for developing a system that will address existing limitations and enhance institutional efficiency.

APPENDIX 1: INTERVIEW QUESTIONS USED IN THE FIELD STUDY.

1. What key features or functionalities would you like to see included in this student management system?
2. What specific challenges are you currently facing with file based approach that this system should solve?
3. Who do you expect will use the system, and what tasks should each user type be able to perform?
4. What types of student information should the system be able to store, track, and report on?
5. Are there any specific reports or data insights you would need from the system on a regular basis?